Cheng Dai

[Email] | [Github] | [Homepage] | [Google Scholar] | [Beijing, China]

Overview

I'm a researcher specializing in world models, video generation and physical AI.

My research investigates how generative models can learn the dynamics, geometry, and physical structure of real-world environments, and how these models can support perception, planning, and interaction.

Positions

Tsinghua University, Institute for AI Industry Research Jul. 2026 - ?

Research Assistant Supervisors: Chaojian Li and Hao Zhao

· Dissertation Fields: Quantum Calculation, World Model, Video Generation, Physical AI

Westlake University, School of Science Sep. 2025 - Jul. 2026

Research Assistant Supervisors: Xin Yuan and Fanglin Bao

· Dissertation Fields: HADAR (nature2023, Temperature-emissivity-texture-distance decomposition)

· AI department manager in the flab. Conduct Research and guide students.

· Release the first thermal hyperspectral image restoration methods based on Radiative Transfer Equation, HAIR.

· Release the first temperauture-emissivity-texture decomposition dataset and benchmark, TeX-1500.

Education

Wuhan University Sep. 2025 - Mar. 2026

Academic Mphil's Student (Withdrawal) in State Key Laboratory of Surveying and Mapping Remote Sensing Information Engineering, Major in Communication and Information Systems *Supervisors: Wei He*

- GPA: 90.79%. First-class entrance scholarship.

Jilin University Sep. 2021 - Jun. 2025

BSc. Major in Communication Engineering

- GPA: 90.58%. Rank: 12/283 (Outstanding graduate, top 5%).
- Outstanding Undergraduate Thesis (top 3%).
- Outstanding Student of the School (*4). First-Class Scholarship, Second-Class Scholarship (*3)
- 508/710 in CET6. Intermediate Russian excellent. One-year German Study.

Publications

[1] Cheng Dai, Jiale Lin, Hongyi Xu, Bingxuan Song, Ziyang Xie, Fanglin Bao

TeX-1500: A Paired Real-World LWIR Hyperspectral Dataset and Benchmark for Temperature-Emissivity-Texture Decomposition. [Paper], [Homepage], [huggingface] (IEEE Transactions on Image Processing under review, IF=13.7)

[2] Cheng Dai, Jiale Lin, Bingxuan Song, Yifei Chen, Jiashuo Chen, Xin Yuan, and Fanglin Bao

HAIR: HADAR-Based Thermal Infrared Hyperspectral Image Restoration. [Paper], [Homepage] (arxiv version)

Intern experience

Lightwheel, Embodied Assets Group Dec. 2024 - Feb. 2025

Group Leader: Jian Yang, and Soroush Nasiriany

- Built tools for embodied simulation, asset construction, data collection, and teleoperation with Blender, MuJoCo, Robosuite, Issacsim and RoboCasa.[Few Contribute Codes, Merged in Final Robocasa]
- Achievement: develop spacemouse, blender2yaml, yaml2blender plugins, reducing validations cycles 20× Faster.

Amoon AI, Large Audio Model Department Apr. 2025 - Sep. 2025

- Independently collect and label 20k infant-cry audios (update 6 versions), and train or finetune models to classify the baby cries and detect activities. Involved Models: BEATs, Audio-MAE, Wav2vec2, Qwen-audio, Whisper, Conformer.
- Achievements: reach 67.2% acc in five-class (diaper, hunger, hug, sleepy, gasping) edge deployment device.

AWARDS IN COMPETITION

- First Prize in the Provincial Level of the National Undergraduate Mathematics Competition。
- Honorable Mention in the American Mathematical Competition。
- Student Leader in Science and Technology Innovation Plan (Changchun Institute of Optics, CAS), **excellent completion**。
- Student Leader in Provincial-level college student innovation and entrepreneurship competition, successful completion.

SKILLS

- **Train and Finetune pretrained Generative Model (Language Model, Vision Language Model, Audio Model .etc.)**
- Improvements to Generative Model (Multi-branch diffusion, Schrodinger Bridge or inference enhancement .etc.) and Interpretability Analysis (i.e. Fisher Information theory, Convex Theory, Derivation of differential equations).
- Model-based Physical Inversion, Numerical Computation, Hyperspectral Image Processing.

Software: [**Cuda-Jax**, Latex, Codex, git, ubuntu, Python, C++, Html5, CSS3, Javascript, MATLAB].